Answer ALL TWENTY FIVE questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 Here are the first four terms of an arithmetic sequence.

1 4 7 10 n

(a) Find an expression, in terms of *n*, for the *n*th term of this sequence.

$T_n = a + (n-1)d$

~~Tn = 1+~~

~~T5 =~~ $\cdot\ T_n = 1 + (n-1)3$

$= 1 + 3n - 3$

$= 3n + 2$

$T_5 = 3 \times 5 + 2$

$= \underline{\underline{17}}$

17

(2)

The *n*th term of a different arithmetic sequence is $5n + 17$

(b) Find the 12th term of this sequence.

$T_{12} = 5 \times 12 + 17$

$= 60 + 17$

$= \underline{\underline{77}}$

77

(1)

(Total for Question 1 is 3 marks)